

Task 5: OOPs – Inheritance and overriding

- Create a child class called **GraduateStudent** that extends your base Student. This allows the child class to automatically reuse the fields, getters, array logic, and validation rules from the parent class without rewriting a single line of that code.
- Add Unique Child Traits: We add a new private attribute unique to graduate students: private String thesisTitle, along with its own public getter (getThesisTitle()).
- introduce an status metric called **qualifiesForHonorRoll()** inside the parent class, which checks if a student has an average of **90.0% or higher**. Then, using **Method Overriding**, the GraduateStudent child class will rewrite this exact method to enforce a stricter rule: graduate students must maintain a **95.0% or higher** to make the honor roll.
- **Parent Method:** We add public boolean qualifiesForHonorRoll() inside the Student class with a base threshold of ≥ 90.0 .
- **Method Overriding:** We copy the exact same method signature inside GraduateStudent, use the **@Override** annotation tag, and change the logic check to ≥ 95.0 .

```
package Student_dataHiding;
import java.util.Scanner;

// — INHERITANCE (CHILD CLASS) —
public class GraduateStudent_inheritance extends Student_dataHiding {
    private String thesisTitle;

    // Child Constructor calling parent super
    public GraduateStudent_inheritance(String rawName, String thesisTitle, Scanner
inputScanner) {
        super(rawName, inputScanner);
        this.thesisTitle = thesisTitle;
    }

    public String getThesisTitle() {
        return this.thesisTitle;
    }

    // — NEW STEP: METHOD OVERRIDING —
    @Override
    public boolean qualifiesForHonorRoll() {
        // Graduate students must hit a tougher threshold (95.0%)
        // Notice it calls the inherited calculateAverage() method seamlessly!
        return calculateAverage() >= 95.0;
    }
}
```

```

import java.util.Scanner;
import Student_dataHiding.GraduateStudent_inheritance;
// — MAIN RUNNER EXECUTIVE SCRIPT —
public class Task5_inheritance {

    public static void main(String[] args) {
        Scanner inputScanner = new Scanner(System.in);

        System.out.println("=== Graduate Student Data Entry ===");
        System.out.print("Enter raw student name (e.g., ' mariah carey '): ");
        String rawName = inputScanner.nextLine();

        System.out.print("Enter Research Thesis Topic: ");
        String thesis = inputScanner.nextLine();

        // Child Object Birth
        GraduateStudent_inheritance activeGrad = new GraduateStudent_inheritance(
            rawName, thesis, inputScanner);

        // --- REPORT GENERATION ---
        System.out.println("\n=====");
        System.out.println("  GRADUATE ACADEMIC REPORT  ");
        System.out.println("=====");
        System.out.println("Student Profile: " + activeGrad.getName());
        System.out.println("Total Classes: " + activeGrad.getGrades().length);
        System.out.println("Research Focus: " + activeGrad.getThesisTitle());

        double avg = activeGrad.calculateAverage();
        System.out.println("Final Average: " + avg + "%");

        // — EVALUATING THE OVERRIDDEN METHOD —
        // Java executes the customized child method logic rules automatically!
        if (activeGrad.qualifiesForHonorRoll()) {
            System.out.println("Honor Roll:  ★ APPROVED (95%+ Target Cleared) ★");
        } else {
            System.out.println("Honor Roll:  Standard Graduate Standing");
        }
        System.out.println("=====");

        inputScanner.close();
    }
}

```